Rethinking the Research Process

How AI Is Changing Research for Students and Professors

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Korea University · AI Literacy for Social Scientists
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Roadmap: AI in research and teaching

1. When to automate, when to protect, when to collaborate?
2. See how knowledge bases and reusable skills support research.
3. Follow a short demo using a recent project of mine.
4. Reflect on takeaways and discuss.

From generative AI to agentic AI

Generative AI	AI that creates new text, images, audio, or code.
Large language model	A model trained on large collections of text to work with and produce language.
Agentic AI	AI that can pursue a task through multiple steps, using tools and responding to results.

This talk focuses on agentic AI, which uses generative models in multi-step workflows to use tools, inspect results, and continue work.

From a conversation to carrying out a task

I mainly use Claude Code and Codex in a CLI.



Terminal	A text-based app for running commands.	CLI	A command-line interface used in a terminal.
Repository	A project folder with files and history.	Tool	An allowed action: search, read, write, or run.

What does it mean to use agents for research?

Software development

Code often has fast, repeatable checks: tests, compilers, or visible outputs.

Research

Research relies on evidence, design, and interpretation. Checks are often slower and less mechanical.

Verifiability

Can we inspect what happened?

Validity

Does the evidence support the conclusion?

Software lesson adapted from Matt Pocock, `diagnosing-bugs` (2026).

An example research claim

"Studies of unilateral policymaking similarly show that citizens penalize executive action relative to legislative action"

Cited together: Reeves & Rogowski (2016); Christenson & Kriner (2017).

An excerpt from an earlier draft of my manuscript.

Do these sources support this claim? Are those even *real* sources?

PART 1

Automate, protect, collaborate

Three modes of AI engagement.

Three modes of AI engagement

Mode	Use it when	Examples
Automate	The work is routine and little is gained by doing it yourself.	Schedule management, filing, formatting.
Protect	Doing the work develops or demonstrates understanding.	Reading, creative thinking. Grading and writing.
Collaborate	AI helps with work you guide and judge.	Verifying sources, data analysis, generating feedback, revisions

AUTOMATE

Cognitively Offload

Efficiency is the goal.

Save time and attention

Schedule	Arrange meetings, maintain calendars, and prepare reminders.
Track	Update task lists, milestones, deadlines, and routine status reports.
File	Name, sort, organize, and convert project materials.
Format	Prepare references, tables, slides, and documents consistently.

PROTECT

Sandbox and protect

Durable learning is the goal.

Research judgment is a practiced capacity

It develops through repeated decisions about evidence.

Interpret

What does this source actually establish?

→

Weigh

How much confidence does the evidence deserve?

→

Argue

What claim can the evidence bear?

When an agent makes these decisions, it can improve the answer while weakening the capacity to judge it.

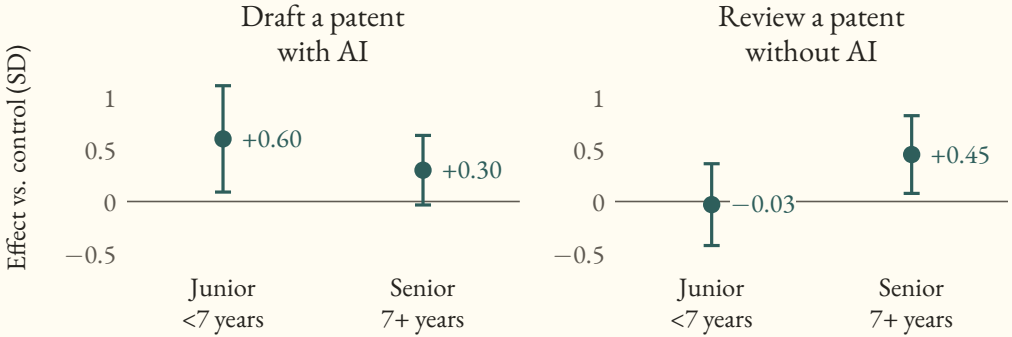
Performance and learning need different evidence

Study	What was assessed	Finding
Autor et al. (2026)	Assisted patent drafting and later unaided review.	Senior lawyers improved. The junior effect was unclear.
Bastani et al. (2025)	Assisted practice and a later unaided mathematics exam.	The tutor reduced exam harm. No gain was established.
Bassner et al. (2026)	Programming exercise and learning assessment.	Exercise scores improved. No learning gain was established.

Autor et al., NBER 35720. Bastani et al., PNAS (2025). Bassner et al., CAI 10, 100537.

AI assistance and later unaided review

Patent lawyers completed a later patent-review task without AI.

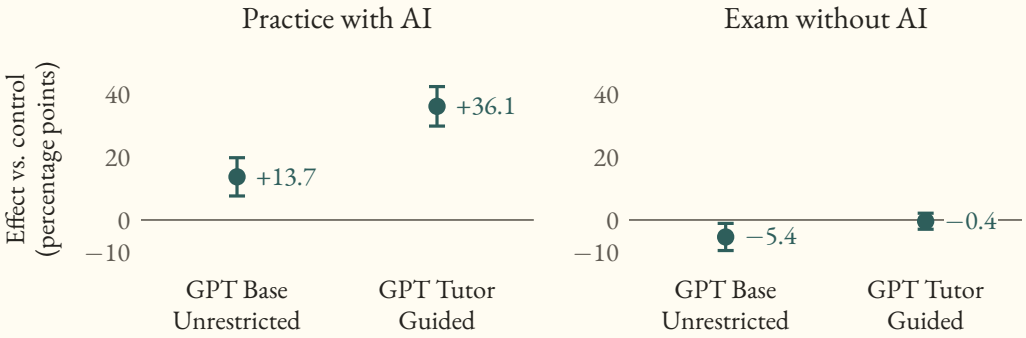


Both groups drafted better with AI, but only senior lawyers improved on unaided review.

Bars show approximate 95% intervals. Autor et al. (2026), NBER 35720, Tables 4 and 6, col. 4.

Practice with AI, then an exam without it

High-school mathematics. Assisted practice was followed by an unaided exam.



Guided AI raised practice scores, but unaided exam performance did not improve.

Bars show approximate 95% intervals. Bastani et al. (2025), PNAS, Table 1. DOI 10.1073/pnas.2422633122

COLLABORATE

Combine human judgment with machine capability

Better, human-machine work is the goal.

Collaboration across research and teaching

Sources

Find literature.
Locate passages.
Compare findings.

Analysis

Organize data.
Write and test code.
Visualize results.

Writing

Structure and revise drafts. Compare claims with sources.

Supervision

Helping augment/structure feedback and guided learning.

The machine expands research possibilities. The researcher/teacher remains responsible for what we do and the output.

What might support learning *with* AI?

Relevant prior knowledge

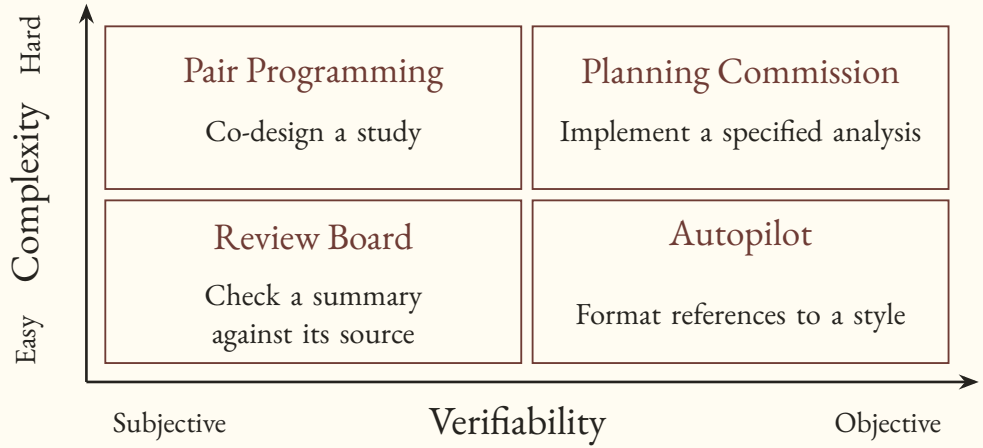
Professional seniority is not the same as familiarity with a new task.

Engagement with the process

Using AI can support learning, but not when it substitutes for engagement with the underlying material.

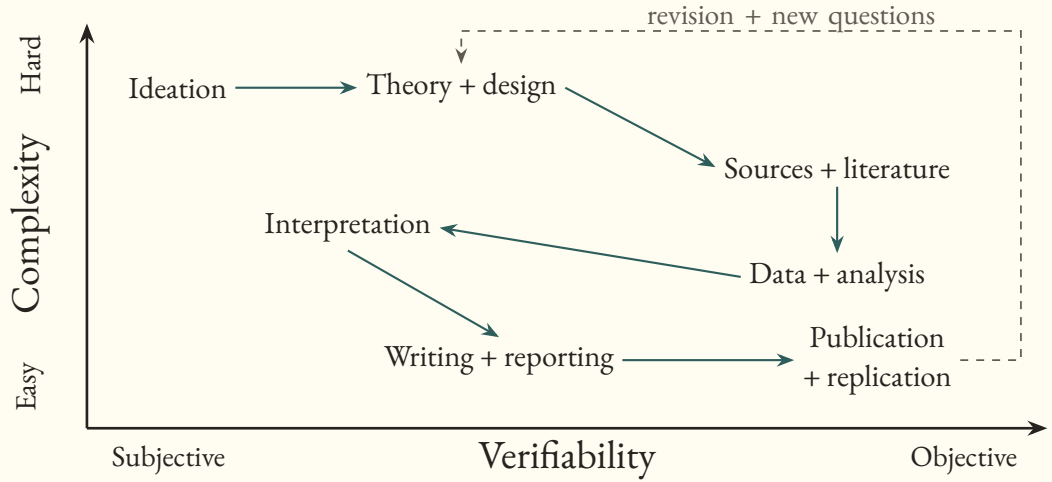
Shen & Tamkin (2026), preprint. Melumad & Yun (2025), PNAS Nexus 4(10).

Match supervision to the research task



Framework from Kenny (2026), p. 16, adapting Swaney's "Claude Overload" workshop. Illustrative research tasks added here.

Research moves through changing risk terrain



Illustrative zones, not scores.

Matching actors to the supervision arrangement

Who plans, who acts, and who reviews?

Arrangement	Plans	Acts	Reviews
Pair programming	Human + agent	Human + agent	Human + agent
Planning commission	Human	Agent	Human
Review board	Agent	Agent	Human
Autopilot	Human instructs	Agent	Agent checks

Kenny (2026), p. 16, adapting Swaney's "Claude Overload" workshop.

PART 2

Building verifiable AI workflows

From knowledge base to replication.

Building a knowledge base (LLM wiki)

Building a structured file system that an agent can search, read, update, and cite.



The files persist after the LLM session ends.

Related to Karpathy's LLM Wiki pattern. See the Astro-Han implementation and MindStudio overview.

A folder or repo is not by default a knowledge base

A new project folder

- Scattered PDFs
- Drafts with unclear provenance
- No shared naming rules

A working knowledge base

- Readable source files
- Stable links to source identity
- Instructions for maintenance

The structure must be created and maintained. A great use of agent “skills.”

Repeated and structured task? Make a skill

A skill is a stored procedure for a task you expect to do again.

Instructions	What to do and what standard to follow.
Resources	Optional scripts, templates, and examples.
Reuse	Invoke it when needed; inspect and improve it over time.

What does a skill actually do?

Without

```
Check whether [claim] is supported.
```

The model and its agents decide how to do the task.

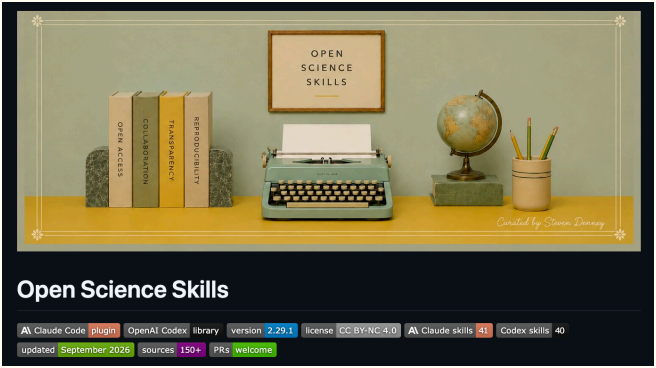
With a skill

```
/oss:fact-check manuscript.tex
```

The procedure specifies the sources, checks, evidence, and report.

A skill makes the procedure more predictable and reusable.

Open Science Skills



A public library of reusable research procedures for Claude Code and Codex.

Open Science Skills provides research-process oriented skills for Claude Code and Codex.

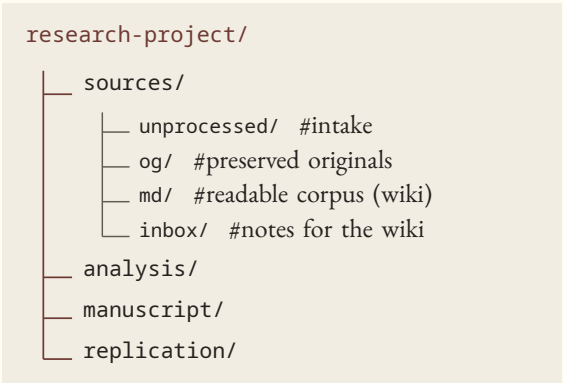
Skills for different parts of the research flow

Knowledge base	research-repo	Preserve and connect the evidence
Source identity	citation-check	Resolve citations and bibliographic records
Claim support	fact-check	Compare manuscript claims with filed sources
Paper and package	paper-review-lite replication-package	Review the paper and execute package checks

Each skill changes the division of planning, action, and review between researcher and agent.

`research-repo` builds the project around your sources

Keep source evidence, analysis, writing, and replication materials in one persistent project.



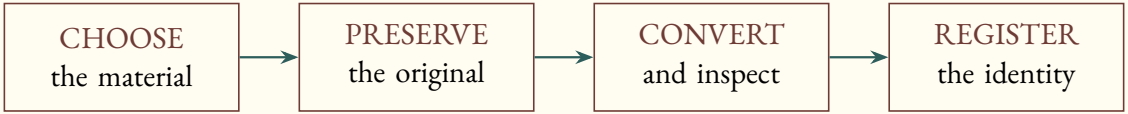
Project repo

Sources, code, drafts, review records, and instructions remain available together.

Project history

Git makes changes recoverable and attributable across sessions. Just use GitHub!

Source collection keeps the researcher “in the loop”



OpenDataLoader PDF creates readable Markdown

Preserved PDF

> **Abstract:** *The formal institutional constraints that Congress and the courts impose on presidential unilateral action are feeble. As a result, recent scholarship suggests that public opinion may be the strongest check against executive overreach. However, little is known about how the public assesses unilateral action.*

Tracked Markdown conversion

```
Abstract. The formal institutional constraints that Congress and the courts impose on
presidential unilateral action are feeble. As a result, recent scholarship suggests
that public opinion may be the strongest check against executive overreach.
```

When a PDF contains figures, extracted image files are relegated to a separate folder with images linked by path.

Git connects the knowledge base to writing in Overleaf

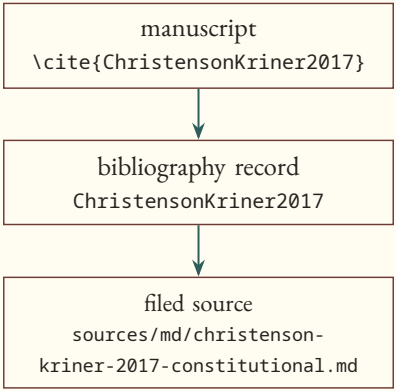


The paper and the evidence base can share a versioned project history.

Overleaf supports Git access for projects with the relevant account access.

BibTeX keeps source identity connected

```
@article{ChristensonKriner2017,
author = {Dino P. Christenson and Douglas L.
Kriner},
title = {Constitutional Qualms or Politics as
Usual?},
journal = {American Journal of Political Science},
year = {2017},
doi = {10.1111/ajps.12262}
}
```



The agent extracts candidate metadata, verifies it, then registers the record after conversion.

A citation can be matched to the wrong work

`citation-check` starts with the keys that actually appear in the manuscript.



MATCH The key, bibliographic record, and resolved metadata must match.

`fact-check` turns prose into a source test

HYPOTHETICAL CASE

MANUSCRIPT CLAIM	FILED SOURCE	VERDICT
“A field experiment increased voter turnout by 12 percentage points” (Author 2025).	“Turnout increased by four percentage points in the treatment group.”	CONTRADICTED The direction matches. The magnitude does not.

A valid citation can still support the wrong number.

Hypothetical example. The report returns the verdict, quoted passage, and suggested correction in the agent session.

The reviewers are reviewed too

`paper-review-lite` checks the manuscript through a verification cascade.

9

specialist reads

argument · numbers · citations
writing · design · figures
preregistration · archive

→

2

cross-checkers

verify quoted passages
remove false positives

→

1

pre-submit report

ranked findings
journal readiness

Every critical or recommended finding must point to a quoted manuscript passage.

Static package verification runs before the agents. Executing the analysis still requires authorization.

`replication-package` prepares a project for deposit

The skill gathers the paper's research materials into a documented replication package.

RESEARCH PROJECT

```
data/
analysis/
figures/
tables/
manuscript/
```

→

1 Find

Materials tied to the paper

2 Build

A separate package structure

3 Document

README, crosswalk, codebook, and license

→

```
replication/

README.md
master.R
data/
code/
docs/
outputs/
LICENSE
```

Researcher reviews the package, then deposits it in Dataverse, OSF, Zenodo, or another repository.

PART 3

Follow the evidence

Two mistakes from my own work

CASE 1

What evidence would support this claim?

ARCHIVED DRAFT · Denney, S. (2026), *Governing Immigration by the Rules*

> "Studies of unilateral policymaking similarly show that citizens penalize executive action relative to legislative action" (Reeves & Rogowski 2016; Christenson & Kriner 2017).

Sources must compare support for the same policy under executive vs. legislative action.



`fact-check` uses the knowledge base to check the claim made in the manuscript.

CASE 1

`fact-check` evaluates each source against the test

The skill retrieves the source, identifies the relevant passage, and tests whether it supports the claim.

Reeves & Rogowski

"Only about a quarter of respondents in any of the surveys supported unilateral policy making"

SUPPORTS Low generalized support

DOES NOT Same-policy route comparison

Christenson & Kriner

The student-loan route "has no significant influence on the probability of a subject backing the president"

SUPPORTS No significant route effect

DOES NOT General executive-route penalty

UNSUPPORTED Neither source supports the comparative claim as written.

CASE 1

The report recommends. The researcher decides.

`fact-check` returns a verdict, quoted evidence, and a suggested revision.

SUGGESTED REVISION

> Replace the general route-penalty claim with narrower statements that match what each source supports.

RESEARCHER REVISION

BEFORE

> Studies of unilateral policymaking similarly show that citizens penalize executive action relative to legislative action `\citep{ReevesRogowski2016,ChristensonKriner2017}`

AFTER

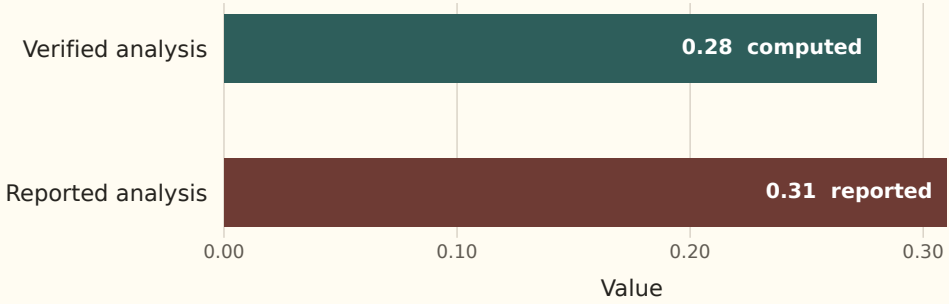
> Studies of unilateral policymaking find that Americans express low generalized support for unilateral powers `\citep{ReevesRogowski2016}` while judging particular unilateral acts largely by whether they agree with them `\citep{ChristensonKriner2017}`

The recommendation is *advisory*. The researcher decides whether and how to revise.

CASE 2

Catching reporting and analysis errors

PRE-SUBMISSION REVIEW · `paper-review-lite` flagged a reported result for verification




```
FAIL    generated 0.28    abstract 0.31    table 0.31
```

Illustrative project-specific comparator triggered by pre-submission review. This checks consistency rather than validity.

CASE 2

The agent flags the mismatch. The researcher decides.

PRE-SUBMISSION REVIEW

1 Detect

Run the comparison and flag both stale fields. →

2 Judge

Inspect the data and measure. Authorize the correction. →

3 Repair

Update both fields and rerun the same check.

The correction is complete only when the same comparison passes.

PASS `generated 0.28` `abstract 0.28` `table 0.28`

Inconsistency detection is highly automatable. Validation remains with the researcher.

Conclusion & Discussion

We’re Still Figuring This Out

My takeaways

Fit	Agentic AI is better suited to controlled environments (e.g., software). Research is more brittle and requires repeated, explicit checks.
Verify	Check sources, claims, analysis, reporting, and replication. Passing one check does not validate the next. Verify repeatedly and with agents checking agents.
Preserve	Keep sources, transformations, code, decisions, and revisions connected and recoverable. LLM wikis/knowledge bases are good for research projects.
Disclose	Document who planned, acted, and reviewed; which models, tools, and skills were used; what they accessed; and what was checked.

Use agents to make research verifiable, inspectable, and transparent.

Resources

AI for Research

scdenney.github.io/ai-for-research

Open Science Skills

github.com/scdenney/open-science-skills

Follow-up on Substack

pixelsandpatterns.substack.com

Subscribe for a forthcoming written follow-up and a linked demonstration of setting up a knowledge base.

APPENDIX

References

References 1 of 3

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